

Ex. No:6 Use SleuthKit to analyse digital evidence

Description

The Sleuth Kit (TSK) is a collection of command-line tools that allow you to analyze disk images and recover digital evidence. Here's a step-by-step guide to using Sleuth Kit on a Windows machine to analyse digital evidence:

Step 1: Install Sleuth Kit

1. Download Sleuth Kit:

- Visit the official [Sleuth Kit website](https://www.sleuthkit.org/) or https://drive.google.com/drive/u/1/folders/1iISFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v
- and download the latest version for Windows.

2. Install Sleuth Kit:

- Run the installer and follow the instructions to install Sleuth Kit on your Windows machine.

Step 2: Acquire the Disk Image

Before analysis, you need a disk image of the evidence. This can be an image of a hard drive, memory card, or any other storage device.

1. Create Disk Image:

- Use a tool like FTK Imager or dd to create a bit-by-bit copy of the storage device. Ensure the image is in a format supported by Sleuth Kit, such as .dd, .raw, .img, or .E01.
- Download the below files from the google drive
 1. 4Dell Latitude CPi.E01
 2. 4Dell Latitude CPi.E02

Step 3: Mount the Disk Image (Optional)

Mounting the disk image makes it easier to analyse the file system.

1. Mount the Image:

- You can use a tool like OSFMount to mount the image as a virtual drive on your Windows system.
- This step is optional but helps with navigating the file system easily.

Step 4: Analyse the File System

Use Sleuth Kit tools to analyse the file system and locate evidence.

1. Navigate to the Sleuth Kit Directory:

- Open the Command Prompt and navigate to the directory where Sleuth Kit is installed.

2. Identify File System Type with fsstat:

- Run the command:

arduino

```
fsstat [image file] > filesystem_info.txt
```

- This command outputs information about the file system, which is crucial for understanding the structure of the disk.

3. List Partitions with mmls:

- Run the command:

arduino

```
mmls [image file] > partitions.txt
```

- This command lists the partitions within the image file.

4. Analyze File System with fls:

- Run the command:

arduino

```
fls -r [image file] > file_list.txt
```

- This command recursively lists files and directories in the file system, showing their metadata.

5. Recover Deleted Files with icat:

- To extract a specific file, use icat:

css

```
icat [image file] [inode number] > [output file]
```

- Replace [inode number] with the inode of the file you want to recover, which you can find from the fls output.

Step 5: Analyze Metadata

Extract metadata from files to understand more about the file's history.

1. View Metadata with istat:

- Run the command:

css

```
istat [image file] [inode number] > metadata_info.txt
```

- This provides detailed information about a file, including timestamps, size, and allocation status.

Step 6: Timeline Analysis (Optional)

Creating a timeline of file activity can be crucial in an investigation.

1. Create Timeline with mactime:

- Generate a body file using fls:

css

```
fls -m / -r [image file] > body.txt
```

- Then create the timeline:

CSS

mactime -b body.txt > timeline.txt

- The timeline includes MAC (Modified, Accessed, Changed) times of files.

Step 7: Generate a Report

Document your findings by compiling all the data into a comprehensive report.

1. Compile the Data:

- Gather all the output files (e.g., filesystem_info.txt, partitions.txt, file_list.txt, metadata_info.txt, timeline.txt).

2. Analyze and Document:

- Review the findings, highlight important evidence, and write a report summarizing the investigation's results.

Step 8: Finalize and Store Evidence

Ensure that all evidence and reports are securely stored.

1. Archive Evidence:

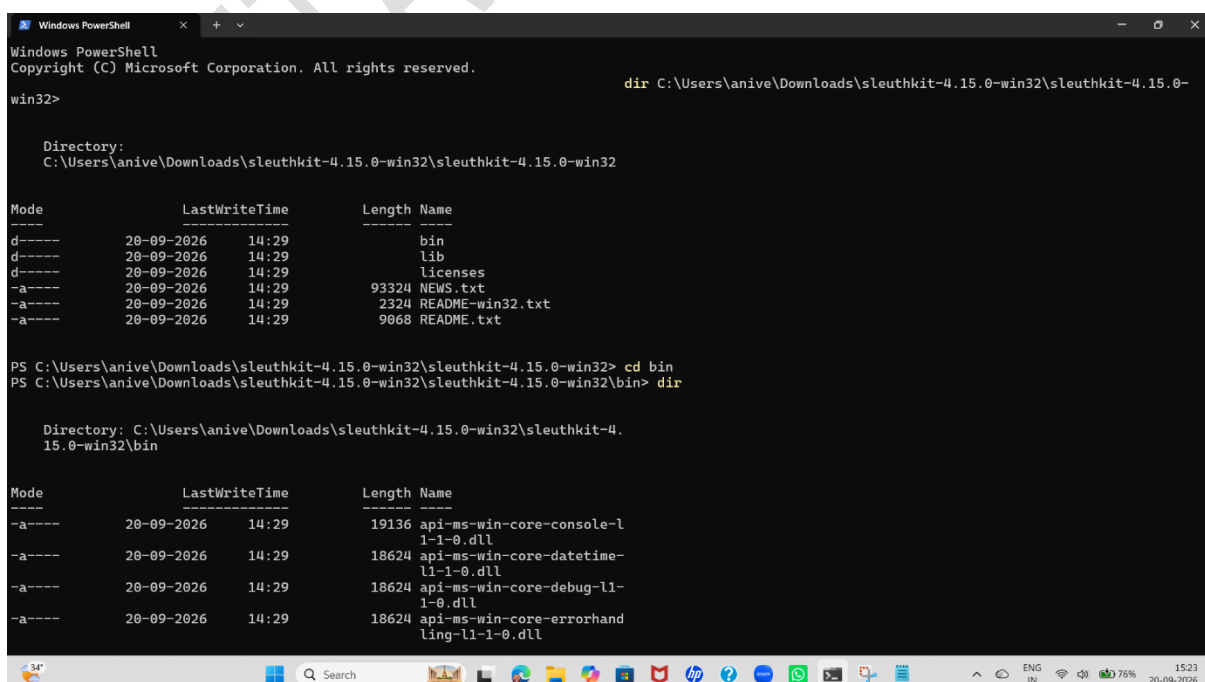
- Use a secure method to archive the disk image and analysis results, ensuring integrity and confidentiality.

2. Store Securely:

- Store the archived data in a secure location, following the chain of custody procedures.

By following these steps, you can use Sleuth Kit on a Windows machine to effectively analyze digital evidence and extract crucial information for your investigation.

OUTPUTS:



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32> dir C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32

Directory:
C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32

Mode                LastWriteTime         Length Name
----                -
d-----         20-09-2026   14:29             bin
d-----         20-09-2026   14:29             lib
d-----         20-09-2026   14:29           licenses
-a-----         20-09-2026   14:29        93324 NEWS.txt
-a-----         20-09-2026   14:29        2324 README-win32.txt
-a-----         20-09-2026   14:29        9068 README.txt

PS C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32> cd bin
PS C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin> dir

Directory: C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin

Mode                LastWriteTime         Length Name
----                -
-a-----         20-09-2026   14:29        19136 api-ms-win-core-console-l1-1-0.dll
-a-----         20-09-2026   14:29        18624 api-ms-win-core-datetime-l1-1-0.dll
-a-----         20-09-2026   14:29        18624 api-ms-win-core-debug-l1-1-0.dll
-a-----         20-09-2026   14:29        18624 api-ms-win-core-errorhandling-l1-1-0.dll
```

```
PS C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin> Move-Item "C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin\"
PS C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin> cd
PS C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin> dir

Directory: C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin

Mode                LastWriteTime         Length Name
----                -
-a----             20-09-2026      14:37         671094597 #Dell Latitude Cpi (1).E01
-a----             20-09-2026      14:39         419384951 #Dell Latitude Cpi (1).E02
-a----             20-09-2026      14:29          19136 api-ms-win-core-console-l1-1-0.dll
-a----             20-09-2026      14:29          18624 api-ms-win-core-datetime-l1-1-0.dll
-a----             20-09-2026      14:29          18624 api-ms-win-core-debug-l1-1-0.dll
-a----             20-09-2026      14:29          18624 api-ms-win-core-errorhandling-l1-1-0.dll
-a----             20-09-2026      14:29          22208 api-ms-win-core-file-l1-1-0.dll
-a----             20-09-2026      14:29          18624 api-ms-win-core-file-l1-2-0.dll
-a----             20-09-2026      14:29          18624 api-ms-win-core-file-l2-1-0.dll
-a----             20-09-2026      14:29          18624 api-ms-win-core-handle-l1-1-0.dll
```

```
PS C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin> .\mmls.exe "4Dell Latitude Cpi (1).E01"
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

Slot      Start      End      Length  Description
000: Meta      0000000000 0000000000 0000000001 Primary Table (#0)
001: -----      0000000000 0000000062 0000000063 Unallocated
002: 000:000 0000000063 0009510479 0009510417 NTFS / exFAT (0x07)
003: -----      0009510480 0009514259 0000003780 Unallocated
PS C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin> .\fsstat.exe -o 63 "4Dell Latitude Cpi (1).E01" > filesystem_info.txt
PS C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin> .\fls.exe -o 63 -r "4Dell Latitude Cpi (1).E01" > file_list.txt
PS C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin> notepad filesystem_info.txt
PS C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin> notepad file_list.txt
PS C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin> .\icat.exe -o 63 "4Dell Latitude Cpi (1).E01" 10091 > wizard_data_recovered.dat
PS C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin> .\listat.exe -o 63 "4Dell Latitude Cpi (1).E01" 10091 > metadata_info.txt
PS C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin> notepad metadata_info.txt
PS C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin> .\icat.exe -o 63 "4Dell Latitude Cpi (1).E01" 11156 > yahoo_recovered.dat
PS C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin> .\listat.exe -o 63 "4Dell Latitude Cpi (1).E01" 11156 > metadata_info2.txt
PS C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin> notepad metadata_info2.txt
PS C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin> .\Rename-Item yahoo_recovered.dat hdr_bluyel_recovered.gif
PS C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin> ii hdr_bluyel_recovered.gif
PS C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin> ii hdr_bluyel_recovered.gif
PS C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin> cmd /c ".\icat.exe -o 63 "4Dell Latitude Cpi (1).E01" 11156 > hdr_bluyel_fixed.gif"
```

```
PS C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin> Get-Item hdr_bluyel_fixed.gif

Directory: C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin

Mode                LastWriteTime         Length Name
----                -
-a----             20-09-2026      15:11          482 hdr_bluyel_fixed.gif

PS C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin> ii hdr_bluyel_fixed.gif
PS C:\Users\anive\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin> |
```

```
File Edit View H1 B I S G D A

FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: B26C1CFCB1B098
OS Name: NTFS
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 2097152
First Cluster of MFT Mirror: 4755208
Size of MFT Entries: 3824 bytes
Size of Index Records: 4096 bytes
Range: 0 - 12305
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 0 - 9520435
Total Sector Range: 0 - 9518415

$AttrDef Attribute Values:
-----
$STANDARD_INFORMATION (16) Size: 40/2 Flags: Resident
$ATTRIBUTE_LIST (32) Size: No Limit Flags: Non-resident
$FILE_NAME (48) Size: 40/479 Flags: Resident
$OBJECT_ID (64) Size: 0-256 Flags: Resident
$SECURITY_DESCRIPTOR (96) Size: No Limit Flags: Non-resident
$VOLUME_NAME (96) Size: 2-256 Flags: Resident
$VOLUME_INFORMATION (112) Size: 12/12 Flags: Resident
$DATA (128) Size: No Limit Flags: Resident
$INDEX_ROOT (144) Size: No Limit Flags: Resident
$INDEX_ALLOCATION (160) Size: No Limit Flags: Non-resident
$BITMAP (176) Size: No Limit Flags: Non-resident
$SPARSE_POINT (192) Size: 0-16384 Flags: Non-resident
$EA_INFORMATION (208) Size: 0-8 Flags: Resident
$EA (224) Size: 0-65536 Flags:
$logGEDUTILITY_STREAM (256) Size: 0-65536 Flags: Non-resident
```

```
File Edit View H1 B I S G D A

d/d 3671-144-7: Documents and Settings
+ d/d 3673-144-7: All Users
++ d/d 3787-144-7: Application Data
+++ r/r 3800-128-1: desktop.ini
+++ d/d 3708-144-7: Microsoft
++++ d/d 3709-144-1: Crypto
+++++ d/d 3712-144-1: OS
+++++ d/d 3713-144-1: MachineKeys
+++++ d/d 3710-144-1: RSA
+++++ d/d 3711-144-1: MachineKeys
++++ d/d 18977-144-1: Dr Watson
++++ r/r 18979-128-3: drvtstn32.log
++++ r/r 18980-128-3: user.dmp
++++ d/d 7325-144-1: HTML Help
++++ d/d 7287-144-7: Media Index
++++ r/r 6874-128-5: wmlibrary_v_0_12.db
++++ d/d 4759-144-1: Network
++++ d/d 4762-144-1: Connections
+++++ d/d 4763-144-1: Pak
+++++ r/r 18982-128-1: raspohne.pak
+++++ r/r 4705-128-4: sharedaccess.ini
++++ d/d 5663-144-7: User Account Pictures
+++++ d/d 5664-144-7: Default Pictures
+++++ r/r 5665-128-4: airplane.bmp
+++++ r/r 5666-128-4: astronaut.bmp
+++++ r/r 5667-128-4: ball.bmp
+++++ r/r 5674-128-4: beach.bmp
+++++ r/r 5668-128-4: butterfly.bmp
+++++ r/r 5675-128-4: car.bmp
+++++ r/r 5669-128-4: cat.bmp
+++++ r/r 5676-128-4: chess.bmp
+++++ r/r 5677-128-4: dirt bike.bmp
+++++ r/r 5678-128-4: dog.bmp
+++++ r/r 5679-128-4: drip.bmp
+++++ r/r 5680-128-4: duck.bmp
+++++ r/r 5670-128-4: fish.bmp
+++++ r/r 5681-128-4: frog.bmp
+++++ r/r 5672-128-4: guitar.bmp
+++++ r/r 5682-128-4: horses.bmp
+++++ r/r 5683-128-4: kick.bmp
+++++ r/r 5684-128-4: lift-off.bmp
+++++ r/r 5685-128-4: palm tree.bmp
+++++ r/r 5671-128-4: pink flower.bmp
+++++ r/r 5686-128-4: red flower.bmp
+++++ r/r 5687-128-4: skater.bmp
+++++ r/r 5673-128-4: snowflake.bmp
+++++ r/r 5688-128-4: guest.bmp
+++++ r/r 5671-128-4: Mr. Evil.bmp
++ d/d 3793-144-1: Desktop
++ d/d 3803-144-7: Documents
+++ r/r 3804-128-1: desktop.ini
```

```
File Edit View H1 B I S G D A

MFT Entry Header Values:
Entry: 11156 Sequence: 2
$logFile Sequence Number: 34998869
Allocated File
Links: 2

$STANDARD_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 295 (5-1-5-21-2000478354-688789844-1708537768-1003)
Created: 2004-08-25 20:55:43.391201600 (India Standard Time)
File Modified: 2004-08-25 20:55:43.391201600 (India Standard Time)
MFT Modified: 2004-08-25 20:55:43.391201600 (India Standard Time)
Accessed: 2004-08-25 20:55:43.391201600 (India Standard Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: HDR_BI-1.GIF
Parent MFT Entry: 8361 Sequence: 3
Allocated Size: 0 Actual Size: 0
Created: 2004-08-25 20:55:43.391201600 (India Standard Time)
File Modified: 2004-08-25 20:55:43.391201600 (India Standard Time)
MFT Modified: 2004-08-25 20:55:43.391201600 (India Standard Time)
Accessed: 2004-08-25 20:55:43.391201600 (India Standard Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: HDR_blow1_u[1].gif
Parent MFT Entry: 8361 Sequence: 3
Allocated Size: 0 Actual Size: 0
Created: 2004-08-25 20:55:43.391201600 (India Standard Time)
File Modified: 2004-08-25 20:55:43.391201600 (India Standard Time)
MFT Modified: 2004-08-25 20:55:43.391201600 (India Standard Time)
Accessed: 2004-08-25 20:55:43.391201600 (India Standard Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0) Name: N/A Resident size: 72
Type: $FILE_NAME (48-3) Name: N/A Resident size: 90
Type: $FILE_NAME (48-2) Name: N/A Resident size: 104
Type: $DATA (128-1) Name: N/A Resident size: 482
```

